
From Records to Reliable Decisions: Measuring and Improving Port Data Maturity

Summary

A port can own advanced software and still lose track of a container. The maturity of its data and analytics system therefore depends on whether people, technology, and governance deliver reliable information across operational handoffs. We develop an auditable framework for Intercontinental Cargo Moving (ICM) Corporation that connects measurement to investment decisions without requiring confidential records to leave the company.

Because the problem provides no corporate observations, we distinguish a deployable assessment system from a reproducible synthetic demonstration. A coverage-constrained screening model selects ten high-frequency indicators from a dictionary of 16, retains mandatory recovery and governance checks, and covers all nine business requirements. The illustrative collection cost falls from 29 to 17 units. Event identifiers, independent audits, missing-data bounds, and versioned rules preserve the link between each score and its evidence.

The Lag-aware Capability Filter (LCF) estimates people, technology, and process capability from monthly audit counts. Its empirical-logit observation model accounts for unequal sample sizes and workload; delayed actions enter the state equation only once. Across 48 simulated months, the test-period capability RMSE is 0.0284, compared with 0.0410 for the direct monthly estimate. One-step prediction RMSE falls from 0.0470 without an onset delay to 0.0349 with validation-selected delays. However, nominal 90% conditional intervals cover only 66.7% of test states. This failure makes uncertainty recalibration a deployment requirement.

The Chain Reliability Maturity (CRM) model translates capabilities into the probability of timely, error-free import and export handoffs. Nonnegative capability effects, a people–process interaction, and a shared-outage term make the score operationally interpretable. In 300 independent synthetic test profiles, the two Brier scores are 0.2289 and 0.2114. The final illustrative capability profile gives a reference maturity of 41.96/100; it is not an estimate of ICM’s actual maturity.

The Robust Improvement Portfolio (RIP) model combines diminishing returns, implementation delays, dependencies, staffing, and quarterly budgets. Exhaustive enumeration compares 16,807 project menus and selects a 5.45-unit portfolio. Its worst planning-scenario net benefit is 5.9671 units, versus 3.5331 for equal allocation. Each unit represents USD 10,000. Under nominal conditions, reference maturity reaches 55.32/100. Wider, held-out uncertainty still produces a 5% budget-violation rate, so the guarantee applies only to the declared planning envelope.

We specify a phased implementation, customer-facing reporting, and transfer rules for smaller ports, larger ports, and trucking companies. Shared identifiers and compatible handoff measures help both port and carrier distinguish the source of delays. Customer confidence rests on traceable evidence, disclosed uncertainty, and corrective action, rather than an unsupported maturity label.

Keywords: data maturity; state estimation; chain reliability; robust optimization; governance; uncertainty

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Reading note. This report solves the 2022 ICM D design problem. Every experiment is synthetic. The supplied 2023 paper is used only as a layout reference. No real team number was provided; the editable number is intentionally unassigned.

1 Introduction: measure the handoff, not the software

1.1 The operational problem

A container's physical journey and its information journey must stay aligned. Arrival notices, manifests, customs releases, yard locations, and inland schedules are produced by different actors. A correct record arriving after a truck has departed can be as costly as an incorrect record. The ICM problem therefore asks for a system that evaluates people, technologies, processes, and their connection [1].

Our assessment covers two complete chains: import, from arrival information to inland release; and export, from inland receipt to vessel loading confirmation. A chain succeeds only when all required information is correct enough for the agreed task and available by its deadline. We do not equate analytics maturity with the number of dashboards, licenses, or sensors.

1.2 Tasks and evidence boundary

1. Define observable KPIs and an interpretable maturity metric.
2. Translate a measured profile into a feasible improvement portfolio.
3. Specify collection, governance, validation, and monitoring protocols.
4. Adapt the system to different port sizes and to trucking firms.
5. Explain these safeguards to customers in a one-page letter.

ICM supplies an operational description but withholds internal records. Consequently, its actual maturity is *not identifiable from the problem statement*. We deliver a measurement instrument and test its behavior using labeled synthetic records. No simulated interview, procurement quote, or corporate financial observation is presented as real evidence.

1.3 Methodological position

Port digitalization connects many organizations, rather than only upgrading equipment [2]. W3C's quality vocabulary separates a quality dimension, its metric, and a measurement [3]; provenance records explain how a result was derived [4]. These ideas motivate our indicator dictionary and audit trail, but supply no numerical port thresholds.

A static weighted score is easier to compute than a state model, but cannot separate noisy observations from delayed capability changes. Bayesian filtering provides a suitable temporal framework [5]. Direct chain-probability modeling avoids multiplying dependent component reliabilities. Proper scoring rules assess whether the resulting probabilities are useful [6]. Scenario robustness then converts them into decisions with explicit uncertainty boundaries [7].

2 Assumptions, notation, and information flow

- A1. Comparable event definitions.** An import/export success has a versioned deadline and error rule. Otherwise probabilities across months are not comparable.
- A2. Anchored capabilities.** One audited pass probability fixes each latent dimension's direction and scale. The demonstration fixes its intercept to zero and loading to one.
- A3. Local response stability.** Within the four-quarter planning horizon, the action-response function is stable. Structural change triggers refitting.
- A4. Audits are conditionally independent in the generator.** Real repeated workers, shifts, or voyages require cluster effects; our reported intervals do not include them.
- A5. Loss is incremental and nonoverlapping.** Avoided losses apply to distinct import/export transactions. Shared-event losses must be allocated once.

Table 1: Core notation. All probabilities and capabilities are dimensionless.

Symbol	Meaning	Unit
$d \in \{P, T, R\}$	People, technology, process	—
t, q, k, j	Month, quarter, chain, project index	—
Y_{dt}, N_{dt}, m_{dt}	Passes, observed audits, unknown records	count
z_{dt}, a_{dt}	Logit capability; $a = \text{logit}^{-1}(z)$	—
p_k, M	Chain success probability; maturity score	—, 0–100
x_j, s_j, G_j	Project level, start quarter, gain ceiling	unit, quarter, —
B_q, Z_s	Quarterly budget; scenario net benefit	USD 10,000

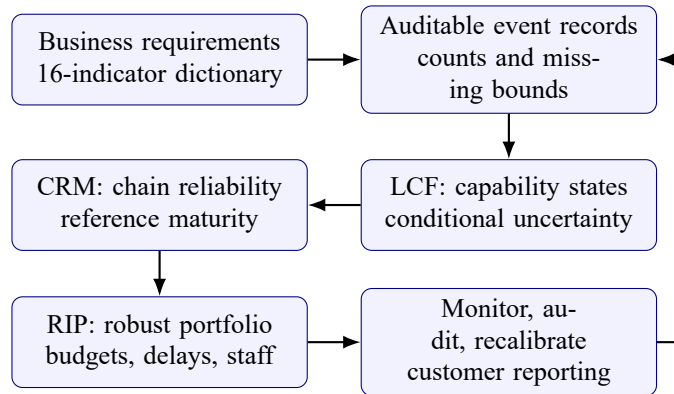


Figure 1: The evidence-to-decision loop. Intervals and source versions accompany each transfer.

3 Indicator dictionary and coverage-constrained screening

Each indicator must answer a business question, be observable, and support an action. A high correlation does not make recovery tests interchangeable with uptime. Table 2 defines the complete dictionary; a star marks the selected high-frequency panel. C2 is an outcome, not an additional independent capability measurement.

Table 2: Operational KPI definitions and management use.

ID	Operational definition	Action or source
P1*	Passed job tasks / valid practical tests	Targeted training; test log
P2*	Qualified staffed skill-slots / required slots	Hire, transfer, or contract
P3	Retained competence after training	Follow-up practical tests
P4	Approval-to-arrival staffing time	Recruitment records; censor unfinished cases
T1*	Available required-service time / required time	Service and outage logs
T2*	Correct delivered logical transactions / due transactions	Interface acceptance tests
T3	Actual minus required data-availability time	Median and tail latency
T4*	Recovery drills meeting requirements / valid drills	Recovery design and rehearsal
R1	Usable required fields / all required fields	Entry and schema rules
R2*	Independently verified correct items / audited items	Source-document audit
R3	Consistent matched records / required matches	Reconcile shared identifiers
R4*	Assets with accountable owners / applicable assets	Ownership register
R5*	Assets meeting metadata rules / checked assets	Catalog and schema version
R6*	Compliant access/lifecycle checks / applicable checks	Approval and disposal evidence
C1	Issue creation to independently confirmed closure	Escalation and root-cause review
C2*	Timely, correct complete chains / due chains	Customer handoff reliability

Let A_{gi} encode coverage of nine requirements: competence, staffing, reliability, interoperability, quality, accountability/security, metadata, lifecycle, and handoffs. With binary inclusion z_i , solve

$$\min_z \sum_i c_i z_i \quad \text{s.t.} \quad \sum_i A_{gi} z_i \geq 1 \quad (\forall g), \quad z_i = 1 \quad (i \in \mathcal{I}_{\text{mandatory}}). \quad (1)$$

The mandatory set is P2, T1, T4, R2, R4, R6, C2. Enumerating 2^{16} subsets selects ten indicators at 17 cost units versus 29 for all indicators, a 41.4% reduction. Raising each cost separately by 20% leaves the set unchanged (minimum Jaccard similarity 1.00). These costs are illustrative effort units. Low-frequency diagnostics remain available; deleting them from the high-frequency panel does not delete the underlying records.

4 Collection protocol and confidentiality

4.1 Collect the denominator as carefully as the numerator

A success rate is meaningless when failed transactions disappear from its denominator. A transaction registry is therefore created before delivery. Every expected handoff receives a business identifier, source and destination, required fields, and a deadline. Retries retain separate technical attempt identifiers but share one business identifier. Audit tests and personnel records use pseudonymous identifiers stored separately from names.

The minimum record is (id, business id, b , event time, ingest time, source, value, quality, version). Event time assigns the reporting month; ingestion time detects lateness. A closed month has a frozen initial report and, where needed, a separately versioned restatement. A missing source record remains visible as an unknown result.

4.2 Stratified manual audits

Logs should be collected exhaustively where practical. Manual tests are stratified by role, shift, terminal, cargo class, and workload. If N_h is the stratum population and y_h/n_h its audited pass rate,

$$\hat{p} = \sum_h W_h \frac{y_h}{n_h}, \quad W_h = \frac{N_h}{\sum_h N_h}, \quad \widehat{\text{Var}}(\hat{p}) = \sum_h W_h^2 \left(1 - \frac{n_h}{N_h}\right) \frac{s_h^2}{n_h}. \quad (2)$$

With per-audit cost c_h and estimated within-stratum standard deviation S_h , continuous optimal allocation satisfies $n_h \propto N_h S_h / \sqrt{c_h}$, subject to a minimum sample in every relevant stratum. Repeated employees or voyages require a cluster bootstrap instead of the independent-record variance above. If a stratum has no responses, report an identification interval rather than a fabricated weighted estimate.

Table 3: Collection risks, corrections, and acceptance evidence.

Risk	Correction	Verification
Night shifts respond less often	Record response by stratum; supplement undercovered shifts	Compare sample and population shares
Confidential records cannot leave ICM	Compute inside ICM; release approved aggregates	Reproduce selected totals from internal evidence
Same auditor defines and validates rules	Freeze rules before independent re-audit	Keep validation sample separate

The demonstration uses one homogeneous audit stratum with 240 expected tests per dimension-month. Stratified reweighting is a deployment protocol, not an experiment claimed to have been completed. A governance owner approves purpose, retention, and access; the evaluation team sees only the information necessary for its task.

5 Processing: preserve errors without rewarding missingness

Raw, cleaned, and analytical layers are separated. The raw layer is append-only. The cleaned layer records every merge, correction, and quarantine reason. The analytical layer supplies counts, uncertainties, applicable populations, and rule versions. Provenance links these entities and transformations [4].

1. **Align meaning and time.** Standardize time zones, event codes, asset classes, and measurement units. Investigate negative delays rather than replacing them with zero.
2. **Deduplicate at the business level.** Exact duplicates share an event identifier. Conflicting duplicates are quarantined for review. Ambiguous cross-system matches retain an unresolved status.
3. **Classify missingness.** Distinguish not applicable, lost logs, late reports, and still-open cases. Open staffing requests and issue durations are right-censored.
4. **Retain real extremes.** A true outage or unusually long customs-related delay is not a data error. Robust flags initiate review, not automatic deletion.

5.1 Partial identification of a pass rate

With N required items, y known passes, and m unknowns, no missingness assumption is needed for

$$p \in \left[\frac{y}{N}, \frac{y+m}{N} \right]. \quad (3)$$

In simulated month 40, the process anchor's observed-only rate is 45.96%, whereas the defensible full-denominator interval is [37.92%,55.42%]. Missing failures are deliberately more frequent in months 40–41. An apparently precise complete-case estimate conceals this uncertainty. We show the filter's failure under that mechanism rather than silently imputing favorable outcomes.

Table 4: Record-conservation audit for the synthetic event file.

Stage	Records
Received records, including duplicates	35,597
Exact duplicates removed	1,037
Unique expected audit events	34,560
Observed outcomes	33,573
Unknown outcomes retained	987
Late flags retained (overlapping category)	920

The equations $35,597 - 1,037 = 34,560$ and $33,573 + 987 = 34,560$ are checked in the reproduction audit. Lateness is only flagged in this synthetic file; receipt-time forecasting is not tested. Monthly train/validation/test blocks are 1–24, 25–36, and 37–48. Parameters and delay choices never use test outcomes.

6 Model I: Lag-aware Capability Filter (LCF)

6.1 From actions to latent capability

For each dimension, let $a_{dt} = \sigma(z_{dt})$, where σ is the logistic function. We use a persistent state with a delayed action input:

$$z_{dt} = \phi_d z_{d,t-1} + (1 - \phi_d) \mu_d + \beta_d \tilde{u}_{dt} + \eta_{dt}, \quad \eta_{dt} \sim N(0, Q), \quad (4)$$

$$\tilde{u}_{dt} = \frac{\sum_{\ell=0}^2 0.6^\ell u_{d,t-\delta_d-\ell}}{1 + 0.6 + 0.6^2}, \quad u_{dt} = 0 \text{ for } t < 1. \quad (5)$$

Here δ_d is onset delay, distinct from the three-month decay window. Persistent action effects accumulate through the state equation. Applying another lag smoother to the same KPI would count the delay twice. The observation mechanism is

$$Y_{dt} \mid z_{dt} \sim \text{Binomial}(N_{dt}, \sigma(z_{dt} - 0.25c_t)), \quad c_t = \sin(2\pi(t-1)/12). \quad (6)$$

The workload coefficient -0.25 and process-noise standard deviation $\sqrt{Q} = 0.08$ are declared scenario assumptions. They are not estimates from ICM. In deployment, estimate them using a pilot and audit-based workload contrasts.

6.2 An explicit working approximation

For computational transparency, we fit an empirical-logit Gaussian working model, not an exact binomial filter:

$$r_{dt} = \log \frac{Y_{dt} + 1/2}{N_{dt} - Y_{dt} + 1/2} + 0.25c_t, \quad (7)$$

$$R_{dt} = \frac{1}{Y_{dt} + 1/2} + \frac{1}{N_{dt} - Y_{dt} + 1/2}, \quad r_{dt} \approx z_{dt} + \epsilon_{dt}. \quad (8)$$

With filtered mean m_{t-1} and variance V_{t-1} , the scalar Kalman update is

$$m_t^- = \phi m_{t-1} + (1 - \phi) \mu + \beta \tilde{u}_t, \quad V_t^- = \phi^2 V_{t-1} + Q, \quad (9)$$

$$K_t = V_t^- / (V_t^- + R_t), \quad m_t = m_t^- + K_t(r_t - m_t^-), \quad V_t = (1 - K_t) V_t^-. \quad (10)$$

Algorithm 1: Estimate and forecast capabilities

Input: monthly counts, workload, dated actions. Fit (ϕ, μ, β) on months 1–24 by minimizing the Gaussian innovation negative log likelihood for each $\delta \in \{0, 1, 2, 3\}$. Select delay on months 25–36. Freeze parameters, then filter and predict months 37–48 sequentially. **Output:** $\sigma(m_t)$, one-step predictions, and conditional intervals $\sigma(m_t \pm 1.645\sqrt{V_t})$.

7 LCF results and its failed uncertainty check

All numerical results on this page are synthetic scenario results.

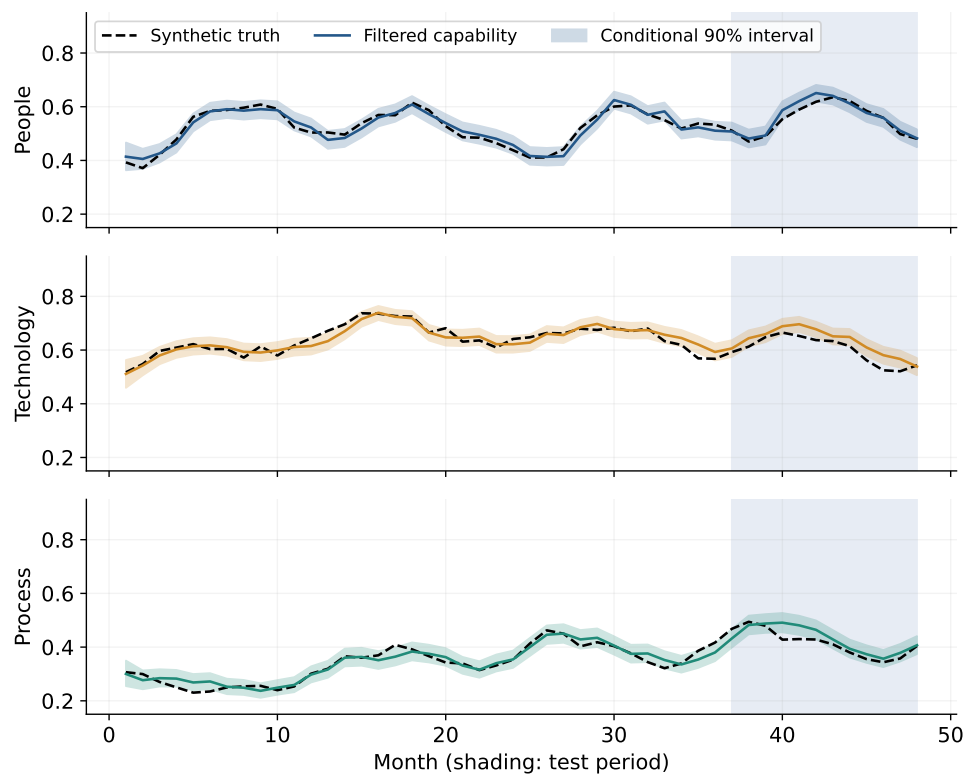


Figure 2: Capability recovery over 48 months. Shading marks the untouched test block; bands condition on fitted parameters.

Table 5: Test-block errors and delay recovery.

Quantity	Result
Filtered capability RMSE / raw monthly RMSE	0.0284 / 0.0410
Delayed one-step RMSE / zero-delay one-step RMSE	0.0349 / 0.0470
Coverage of nominal 90% conditional intervals	66.7%
Selected delays (P,T,R); generating delays	(2, 3, 2); (2, 1, 2)

Filtering reduces sampling noise, and validation-selected onset delays improve the pooled forecast error. Yet technology's selected delay is three months instead of the generating one month. A short, periodic action series does not reliably identify a delay: persistence, baseline, and action strength can compensate for each other. We therefore interpret the model as a forecasting instrument, not proof of a training or software intervention's causal effect.

Only 24 of 36 test states lie inside nominal 90% conditional intervals. Failure-dependent missingness, parameter uncertainty, and the Gaussian approximation all contribute plausible explanations; this experiment does not isolate their shares. Before operational scoring, use independent failure audits, full-denominator bounds, and a validation-calibrated predictive interval. Until coverage is acceptable on fresh data, publish the point estimate with an uncertainty warning rather than a confident certification.

8 Model II: Chain Reliability Maturity (CRM)

Maturity should answer a concrete question: given a defined service environment, how likely is the data system to complete the customer's handoff correctly and on time? A chain-level model measures the joint event directly. It does not multiply stage success rates as if shared platforms were independent.

$$\text{logit } p_k(a, c, o) = b_{0k} + b_{Pk}a_P + b_{Tk}a_T + b_{Rk}a_R + h_k a_P a_R + \gamma_k c + \psi_k o. \quad (11)$$

The people-process interaction represents skilled staff being able to use an assigned owner, consistent meta-data, and an escalation path. o indicates a shared outage, so both chains can deteriorate together. Constrain $b_{Pk}, b_{Tk}, b_{Rk}, h_k \geq 0$ and $\gamma_k, \psi_k \leq 0$. Then

$$\frac{\partial p_k}{\partial a_P} = p_k(1 - p_k)(b_{Pk} + h_k a_R) \geq 0, \quad \frac{\partial p_k}{\partial a_R} = p_k(1 - p_k)(b_{Rk} + h_k a_P) \geq 0. \quad (12)$$

These inequalities prevent an estimated association from recommending deliberately worse skills or governance within the model domain. They are structural restrictions, not empirical proof that every real intervention helps.

8.1 Separate comparable capability from current exposure

$$M^{\text{ref}}(a) = 100 \sum_{k=1}^2 \omega_k p_k(a, 0, 0), \quad M_t^{\text{op}} = 100 \sum_{k=1}^2 \omega_k p_k(a_t, c_t, o_t), \quad \omega = (0.6, 0.4). \quad (13)$$

Reference maturity fixes workload and outage conditions. Operational performance retains the actual environment and must be reported alongside it. The demonstration weights follow an assumed import/export transaction mix, not loss severity; monetary loss weights appear separately in optimization. During a real outage, a stable reference score must never conceal an operational red alert.

We report a continuous score and three-dimensional capability profile. No supported level boundaries are supplied by ICM, so we do not invent five maturity grades. A future certified level would require both an empirically calibrated score threshold and mandatory security, recovery, ownership, and lifecycle checks. A high average cannot offset a failed critical gate. NIST's historical 2018 framework informs the risk-management structure, not a numeric maturity conversion [8].

Algorithm 2: Fit and assess chain reliability

Input: independent chain outcomes, capability profiles, workload, outage flags. Fit constrained logistic regressions on 1,200 profiles; choose ridge penalty from $\{0, 0.001, 0.01\}$ using 300 validation profiles. Freeze the coefficients, score 300 test profiles, and calculate Brier score, log loss, and calibration bins. **Output:** reference/operational scores, capability profile, and critical-gate status.

9 CRM calibration and uncertainty transfer

All numerical results on this page are synthetic scenario results. The pilot generator draws 1,800 independent capability profiles uniformly over $[0.2, 0.9]^3$, workload over $[-1, 1]$, and an outage indicator with probability 0.15. It generates two Bernoulli outcomes per profile from specified coefficients. Broad profiles are needed because one port's short history cannot identify all interactions. These profiles are experimental test cases, not a survey of 1,800 ports.

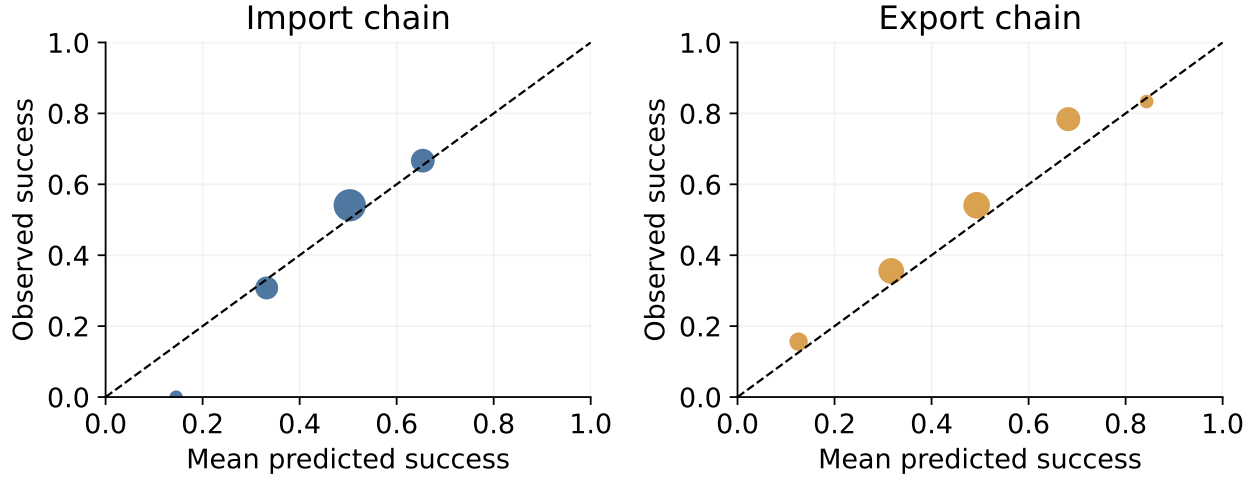


Figure 3: Held-out calibration. Marker size reflects bin count; the dashed line is ideal calibration.

Table 6: Independent synthetic test performance ($n = 300$ per chain).

Chain	Brier	Constant baseline	Log loss	True- p RMSE
Import	0.2289	0.2506	0.6480	0.0742
Export	0.2114	0.2549	0.6117	0.0294

Brier score is $n^{-1} \sum_i (p_i - y_i)^2$; the baseline uses the training success frequency. Both fitted chains improve on that baseline. The import model's stronger regularization shrinks coefficients and gives a larger true-probability error. Calibration bins are descriptive, with unequal sample sizes; they do not establish cross-port validity. Proper scoring rules discourage confident but inaccurate probabilities [6]. The final filtered profile is $(0.4823, 0.5384, 0.4070)$, giving reference maturity 41.96. We propagate 1,000 draws of the final Gaussian state and coefficient vectors based on the observed-information approximation. Within-coefficient covariances are retained; signs are projected onto the permitted domain. The resulting 5th–95th percentile range is $[39.50, 52.71]$. It is an approximate uncertainty envelope, not a calibrated 90% confidence interval. State-parameter uncertainty, shrinkage bias, and cross-chain coefficient dependence remain omitted.

Two whole joint draws nearest the 5th and 95th score percentiles are forwarded to portfolio stress testing. Selecting a few representative draws is tractable, but does not guarantee coverage of every adverse parameter combination.

10 Model III: Robust Improvement Portfolio (RIP)

10.1 Response with delay and diminishing returns

Let project j receive level $x_j \in \{0, 1, 2, 3\}$, with one-time start $s_j \in \{1, 2\}$ when active. Its response at quarter q is

$$f_{jq}^{(s)} = \min \left\{ 1, \max \left(0, \frac{q - s_j - d_j - \Delta_s + 1}{2} \right) \right\}, \quad E_{jq}^{(s)} = x_j f_{jq}^{(s)}, \quad (14)$$

$$g_j(E) = G_j^{(s)}(1 - e^{-k_j E}), \quad g'_j > 0, \quad g''_j < 0, \quad (15)$$

$$a_q^{(s)}(x) = \text{clip} \left(a^{0,(s)} + \sum_j m_j g_j(E_{jq}^{(s)}), 0, 0.995 \right). \quad (16)$$

The capability increment is measured against the same baseline a^0 at every quarter. Adding the total cumulative gain repeatedly to the preceding quarter would create artificial growth. An equivalent recursive implementation must add only $g_j(E_{jq}) - g_j(E_{j,q-1})$.

10.2 Monetary objective and feasibility

$$Z_s(x) = \sum_{q=1}^4 \frac{\sum_k V_{kq} L_{kq}^{(s)} [p_k(a_q^{(s)}(x), 0, 0) - p_k(a^{0,(s)}, 0, 0)]}{(1+r)^{q-1}} - \sum_{q=1}^2 \frac{C_q^{(s)}(x)}{(1+r)^{q-1}}. \quad (17)$$

Here V is the quarterly count and L the loss per failed handoff; their product is converted to USD 10,000 units. Safety and legal obligations are not traded against this monetary benefit. With $y_j = \mathbf{1}(x_j > 0)$, solve

$$\max_{x,s,v} v, \quad v \leq Z_s(x) \quad (\forall s \in \mathcal{S}), \quad (18)$$

$$\sum_{j:s_j=q} \kappa_s(x_j + 0.15y_j) \leq B_q, \quad \sum_{j:s_j=q} h_j x_j \leq H_q, \quad (19)$$

$$y_{\text{integration}} \leq y_{\text{governance}}, \quad s_{\text{governance}} \leq s_{\text{integration}} \quad \text{if integration active}, \quad (20)$$

$$x_{\text{recovery}} \geq 1. \quad (21)$$

A recovery floor is an explicit scenario management decision, not a universal standard. Governance produces an initial usable foundation in its start quarter, so simultaneous integration is allowed in the demonstration. If real acceptance must precede integration, replace this with a strict commissioning dependency. The staffing constraint is a capacity proxy for workload and disruption; no separate downtime guarantee is claimed.

11 RIP parameters and the selected implementation menu

All numerical results on this page are synthetic scenario results.

Table 7: Declared project-response assumptions. Costs are USD 10,000 units.

Project	G_j	k_j	d_j	h_j	$m_j = (P, T, R)$
Training	.16	.80	1	1.2	(1, 0, 0)
Hiring	.18	.55	2	.7	(1, 0, 0)
Integration	.22	.60	1	1.5	(0, 1, 0)
Governance	.24	.75	0	1.0	(0, 0, 1)
Recovery	.12	.65	1	.8	(0, .65, .35)

Budgets are (7, 4) units in the first two quarters; staff capacities are (8, 6) units. The fixed start cost is 0.15 per active project, and the quarterly discount rate is 2%. Each quarter has 6,000 import and 4,000 export handoffs with failure losses of USD 40 and USD 60. These are test assumptions, not quotations or industry benchmarks. The model can leave funds unused when additional projects reduce worst-case net benefit.

Table 8: Planning scenarios. Joint uncertainty adds two complete sampled state/coefficient configurations.

Scenario	Cost	Gain	Extra delay	Value
Nominal	1.00	1.00	0	1.00
Overrun	1.20	.90	0	1.00
Delay	1.10	.80	1	1.00
Lower avoided loss	1.05	.85	0	.80
Joint draw, low / high score	1.10	.85	0	.90

Algorithm 3: Exact search over the declared menu

Input: five projects, seven options per project (off or three levels at either start), six scenarios. Enumerate $7^5 = 16,807$ menus; check budgets under 20% cost overrun, staff capacity, dependency, and recovery floor. Evaluate all scenario objectives for every feasible menu. Choose the largest minimum net benefit. **Output:** start dates, levels, spend, staffing, and a zero optimality gap *within this finite menu*.

Table 9: Selected project menu; inactive projects have no start date.

Project	Units	Start quarter	Cost
Training	1	1	1.15
Hiring	0	–	0.00
Integration	0	–	0.00
Governance	3	1	3.15
Recovery	1	1	1.15

The selected portfolio spends 5.45 units (USD 54,500) in quarter 1, requiring five capacity units. A 20% overrun gives 6.54 cost units, below the seven-unit budget. The nominal final maturity is 55.32, a 13.36-point increase. This menu favors governance, training, and required recovery under the assumed horizon and returns; zero hiring or integration is not a recommendation to eliminate those functions.

12 Decision validation and the price of robustness

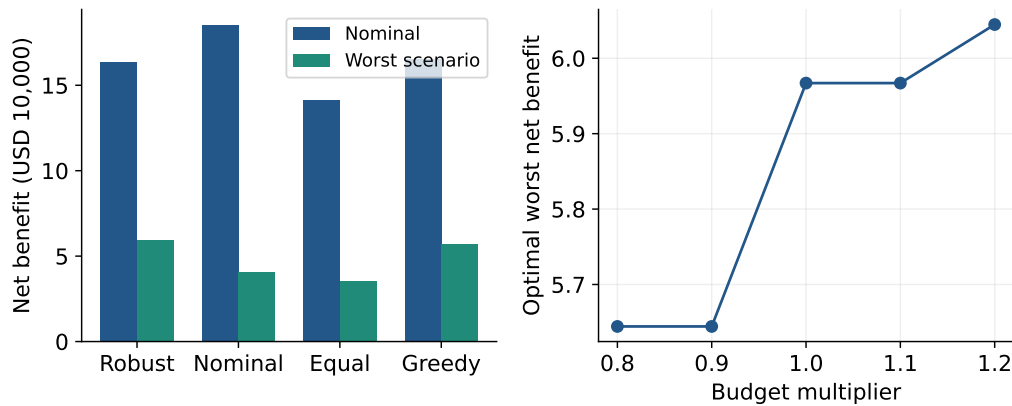


Figure 4: Portfolio performance and reoptimized budget sensitivity. Net benefits are synthetic USD 10,000 units.

Equal allocation gives each project one unit in quarter 1. Greedy search starts from mandatory recovery and adds the feasible one-level increment with greatest nominal gain, also in quarter 1. Nominal optimization uses nominal budgets and returns; robust optimization uses every planning scenario and the 20% cost envelope.

Table 10: Finite-menu comparison in USD 10,000 units.

Plan	Nominal net	Worst net	Q1 spend
Robust	16.3697	5.9671	5.45
Nominal	18.5126	4.0743	6.45
Equal	14.1157	3.5331	5.75
Greedy	16.5363	5.6919	5.60

12.1 Held-out joint shocks

In 200 held-out scenarios, severity $u \sim U(0, 1)$ changes cost to $1 + 0.3u$ and effects to $1 - 0.3u$. Extra delay has probability 0.3; value is uniform on $[0.75, 1.15]$. Half use $G \min(kE, 0.85)$ instead of exponential response. This wider envelope tests both parameter and response-shape errors.

Table 11: Wider held-out scenarios: budget reliability and feasible regret.

Plan	Mean net	Violation rate	Mean feasible regret
Equal	9.542	32.5%	2.146
Greedy	11.001	19.0%	0.478
Nominal	9.760	69.0%	1.312
Robust	10.126	5.0%	1.253

Robust planning reduces violations from 69% to 5%, but greedy search earns more on average with greater budget risk. Regret compares only feasible cases with each scenario's exact optimum; infeasibility is reported separately. Mean return excludes financing and emergency penalties. The 5% rate is a stress-test observation, not a chance-constraint guarantee. Procurement decisions need contingency reserves or a wider cost envelope.

13 Seven scenario families and structural stress tests

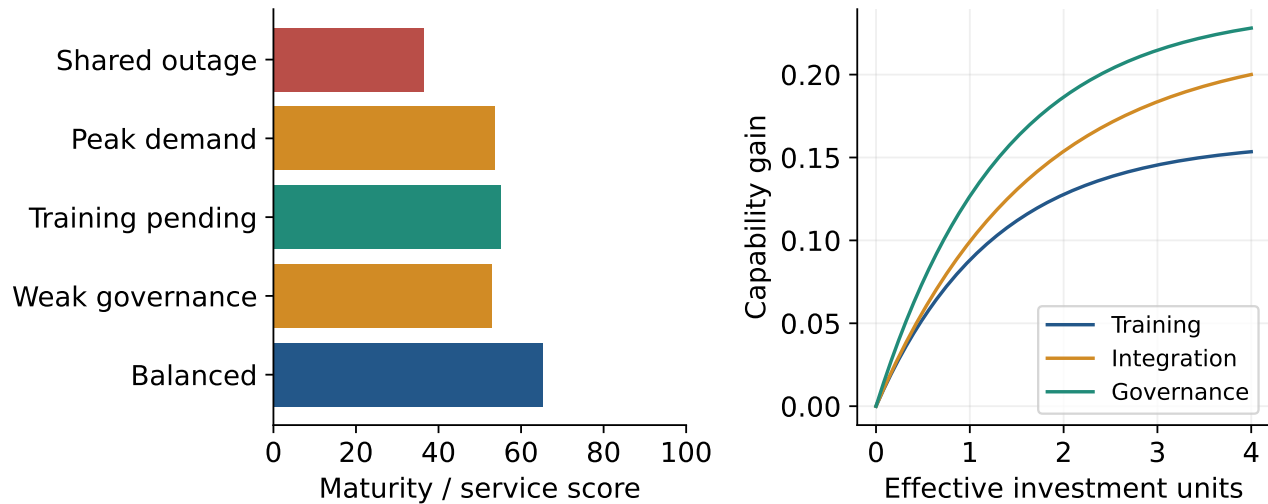


Figure 5: Capability/operational contrasts and diminishing returns. Outage and peak-demand scores use actual conditions.

The first three bars are reference-condition comparisons; the last two are operational stress scores for the balanced profile. They must not be interpreted as five standardized rankings of different ports.

Table 12: Scenario questions and observed model behavior.

Scenario and test	Result and implication
1 Balanced (.7, .7, .7)	65.32 reference points; comparison anchor
2 Strong technology, weak process (.7, .9, .3)	52.82; better hardware does not fully offset governance gaps
3 Training not yet effective (.45, .7, .7)	55.15; separately validate delayed forecasts
4 Peak workload $c = 1$	Operational score 53.69; reference score stays 65.32
5 Shared outage $o = 1$	Operational score 36.45; both chains deteriorate
6 Increasing training investment	First-unit gain .0881, third-unit gain .0178
7 Failure-selective missing logs	Month 40 process interval [37.92%, 55.42%]

The shared-outage test is especially useful: reference maturity isolates capability for comparison, but service reliability can collapse immediately. A customer dashboard should therefore display current incident status before the reference score. The weak-governance comparison also explains why our portfolio does not allocate all resources to technology even when technical products are easier to purchase.

These are designed contrasts. They test whether the model responds in the intended direction, not whether the corresponding score levels are realistic for a specific port. The model would require external outcome calibration before those values support a contract or public assurance claim.

14 Sensitivity, identification, and evidence ledger

14.1 What changes a decision?

A decision is more sensitive than a smooth score when discrete project levels cross budget boundaries. At 80% or 90% of the original budgets, the robust portfolio changes from (1, 0, 0, 3, 1) to (1, 0, 0, 2, 1), and worst net benefit falls to 5.6443 units. A 10% budget increase does not change the menu; a 20% increase adds one training unit and raises worst net benefit to 6.0447 units. Budget expansion therefore has a discontinuous, diminishing decision value.

Table 13: Sensitivity to the assumed state-noise standard deviation; other fitted parameters are frozen.

Change	$\sqrt{Q}$	Test RMSE	Final reference score
−20%	.064	.0290	42.38
0%	.080	.0284	41.96
+20%	.096	.0285	41.63

Small score changes under this one perturbation do not repair the failed interval-coverage test. Sensitivity and calibration answer different questions. The optimization's exponential/piecewise comparison is likewise a structural stress test, not an estimate of real response curvature.

14.2 Evidence ledger and calibration priorities

Table 14: Sources and what each can legitimately support.

Evidence class	Content used here	Required before deployment
Problem facts	Port workflow; confidentiality; P/T/R requirements	Confirm local scope and deadlines
Public references	Quality/provenance concepts; filtering; scoring; robustness	Check applicability and version
Scenario inputs	Costs, staffing, response curves, loss, generator coefficients	Replace with quotes, pilots, finance records
Fitted synthetic parameters	LCF persistence and delays; CRM coefficients	Refit on internal histories and independent outcomes
Management decisions	Reference mix, recovery floor, risk envelope	Document owner approval and rationale

The first pilot should prioritize reliable failure denominators and recovery evidence. Next, obtain varied action histories to separate onset delay from persistence. Only then calibrate money-valued response curves for procurement. Each parameter record stores its unit, source field, effective date, method, uncertainty range, approver, and version. No expert panel was conducted in this work.

15 Turning the score into people and technology decisions

15.1 Staff the missing skill, not an abstract headcount

Build a role–skill–shift matrix for data engineering, analytics, data stewardship, interface operations, and incident response. P2 measures coverage of required cells. P1 and delayed P3 tests establish whether named staff can perform the task, rather than merely complete a course. For skill r and shift h , calculate

$$\text{FTEgap}_{rh} = \max \left\{ 0, \frac{\text{required task hours}_{rh} - \text{qualified available hours}_{rh}}{\text{productive hours per FTE}_h} \right\}. \quad (22)$$

Round and schedule only after accounting for simultaneous shift requirements. A permanent, recurring gap supports hiring; a temporary specialist need supports contracting; an adjacent skill gap supports training when release time and retention are credible. Mixed staffing is often feasible, but its cost, lead time, turnover, and knowledge-transfer obligations belong in the project menu. No defensible number of hires can be calculated from the problem's withheld staffing data.

15.2 Technology selection as a repeatable acceptance process

The model does not recommend a software brand. It compares candidate architectures on common business acceptance tests: a correct logical message after retries; a recovered service within its approved recovery target; interoperable identities; controlled access; lineage; exportable metadata; and latency at peak workload. T1–T4 measure these outcomes at the service level.

Table 15: Architecture comparisons the same model can support.

Question	Single suite	Multiple products
Integration effort	Test internal semantic consistency	Test cross-vendor interfaces and identities
Common failure	Model shared outage concentration	Model shared network/authentication dependencies
Exit and migration	Test full export and restore	Test data portability across components
Economics	License, migration, operations, training	Same costs plus interface maintenance

Start a controlled pilot on one handoff, compare against the current service using the same workload mix, and preserve a rollback path. A vendor's claimed accuracy is not interchangeable with an independently audited C2 rate. Only a verified improvement and a feasible resource plan justify wider rollout.

16 Governance, monitoring, and a phased implementation

Governance determines who can define, change, use, and retire a record. Every critical dataset needs a business owner, a technical steward, an approved purpose, a schema, quality rules, retention conditions, and a documented access path. The owner accepts business risk; the steward implements controls; an independent reviewer checks evidence. These responsibilities are not replaced by the maturity score.

Table 16: Monitoring and response protocol. Thresholds come from approved service commitments.

Frequency	Measure	Owner	Response
Daily	Availability, late critical handoffs, missing logs, security events	Operations and security	Incident triage; preserve evidence; notify affected users
Monthly	KPIs, missing bounds, skills, issue closure, C2	Data owners and HR	Publish versioned report and corrective actions
Quarterly	Calibration, capability drift, portfolio delivery	Model owner and finance	Revalidate; update constraints and budgets
Annual	Indicator scope, access, retention, recovery	Governance committee	Independent review and redesign where needed

An incident, a new data schema, a changed contract, a system migration, or calibration deterioration triggers an extra review. Do not wait for the annual cycle. An illustrative review rule is two successive monthly calibration errors exceeding an internally approved tolerance; the tolerance must be calibrated for sample size rather than chosen to make alerts disappear.

16.1 A 90-day setup followed by quarterly learning

1. **Days 1–30: establish scope.** Inventory the two chains, critical fields, owners, and service commitments. Create expected-event registries and access permissions.
2. **Days 31–60: run a measurement pilot.** Compare source documents with logs, quantify unknowns, test recovery, and identify uncovered skill-slots.
3. **Days 61–90: approve provisional decisions.** Publish an ungraded baseline with uncertainty; gather vendor quotes and staffing costs; select a reversible pilot portfolio.
4. **After sufficient history: fit dynamic models.** The 48-month experiment does not imply a 90-day pilot can estimate all delays. Use direct KPI bounds until enough varied observations accumulate.

The customer dashboard shows current incident status, measured handoff reliability, uncertainty, reporting coverage, unresolved issues, and the date of independent review. Commercially sensitive personnel and financial fields remain internal.

17 Transfer across port sizes and to trucking

17.1 Size-neutral meaning, locally calibrated measurement

A small port should not be penalized for lacking technology it does not need. Retain the people–technology–process structure, denominator rules, and chain-success definition, but recalibrate required service periods, staffing matrices, deadlines, workload controls, and costs. A small port may collect monthly audit samples and use a simple catalog; a large port may need continuous event feeds, segmented ownership, and shared-platform risk analysis. Both must demonstrate the same underlying ability to deliver agreed information. For cargo/shift strata h , standardize to an agreed reference mix:

$$p_b^{\text{std}} = \sum_h \pi_h^{\text{ref}} p_{bh}, \quad \sum_h \pi_h^{\text{ref}} = 1. \quad (23)$$

For example, two equally capable ports have 0.95 success on routine tasks and 0.75 on complex tasks. A 90/10 task mix gives 0.93, whereas a 50/50 mix gives 0.85. At a common 70/30 reference mix, both score 0.89. The eight-point raw difference is workload composition, not evidence of a capability gap. This arithmetic illustration shows why standardization is needed; it does not estimate any real port.

17.2 A trucking company’s compatible metric

Table 17: Mapping the port framework to trucking.

Port element	Trucking counterpart	Common test
Arrival and inland schedule	Order, dispatch, appointment	Timely consistent identifiers
Yard location and release	Vehicle location and pickup authority	Current location and authorized release
Vessel loading confirmation	Proof of delivery	Complete, verifiable closing record
Port data roles	Dispatcher, driver, analyst, steward	Role-specific practical competence

The truck chain is order receipt → dispatch → location/ETA update → pickup or delivery → confirmed receipt. Refit its response coefficients and deadlines; do not copy the port’s 0.6/0.4 weights or score thresholds. Sharing the metric benefits ICM through better arrival planning, fewer conflicting status records, faster attribution of data errors, and a common basis for service reviews.

At the port–carrier boundary, both parties confirm the same handoff identifier and timestamps. Joint success must be measured from matched records: multiplying separately reported port and carrier success rates assumes independence and can conceal common platform failures. Each party retains its own sensitive records while sharing authorized exception summaries.

18 Evaluation, limitations, and recommendations

18.1 What the framework achieves

The measurement dictionary connects nine business needs to inspectable fields and corrective actions. The coverage model reduces illustrative high-frequency collection effort while preserving mandatory controls. Temporal capability estimates retain a distinction between a current observation and a persistent state; chain probabilities translate that state into a customer-relevant service event. The project optimizer accounts for diminishing returns and implementation feasibility, and its finite-menu optimum can be independently verified.

The strongest feature is traceability. A reported score can be followed backward to its chain probabilities, capability estimates, audit counts, missingness decisions, and source definitions. A proposed investment can be followed forward to the conditions under which it is expected to improve the service.

18.2 Limits that materially affect use

1. **No enterprise validation.** All numerical parameters and results come from labeled simulations. The framework cannot assign ICM a genuine maturity score or promise a return on investment.
2. **State uncertainty is undercovered.** Conditional intervals cover only 66.7% of test states. Treating them as calibrated 90% intervals would be misleading.
3. **Some parameters are not identified reliably.** Technology onset delay is misrecovered; coefficient shrinkage alters response curvature. More varied pilots and independent control observations are needed.
4. **Menu optimality is limited.** Only five projects, three levels, and two start quarters are searched. A zero enumeration gap is not global optimality over continuous budgets or every possible project.
5. **Robustness is bounded.** Costs beyond the planning envelope cause 5% violations in the wider test. Legal requirements, catastrophic losses, and multi-year capability decay are not monetized in this demonstration.

18.3 Recommended decision sequence

ICM should first establish expected-event denominators, independent failure checks, ownership, and recovery evidence. It should then publish a provisional capability profile with missing-data bounds and current incident status. After collecting enough varied history, it can calibrate the chain probabilities and run the portfolio model with its own costs, capacity, and service requirements.

In the declared demonstration, a 5.45-unit portfolio raises nominal reference maturity from 41.96 to 55.32. The actionable conclusion is to measure governance and delayed skill effects before committing to a large technology purchase. The numerical portfolio remains conditional on assumptions that have not been verified at ICM.

19 Letter to ICM Corporation's customers

To: ICM Corporation's port users

From: The D&A assessment consulting team

Subject: Making your cargo information more reliable and accountable

Date: September 19, 2026; retrospective study of the 2022 case

Dear Port Users,

You depend on accurate arrival information, customs status, container locations, and pickup schedules. A delay or contradiction in one of these records can disrupt your own operations. We propose a measurement and improvement program that makes these failures visible and gives ICM a clear process for correcting them.

The program assesses whether employees have the skills needed for their jobs, whether technology delivers information when required, and whether data have accountable owners and controlled access. It also checks the complete handoff, from the first notice to the final confirmed transfer. A system will not receive credit simply because it has purchased new software.

Each expected handoff will be counted, including those with missing or late records. Independent checks will compare selected records with their underlying evidence. Reports will distinguish verified results from unknown outcomes, show current service incidents, and explain when a measurement has been revised. This allows you to understand both performance and the limits of the available evidence.

Your information will be used only for approved purposes. Access responsibilities, data retention, and authorized sharing will be documented. Confidential employee details and commercial records will remain within the appropriate organization. Public or customer reports will use approved summaries while preserving an internal audit trail.

When the evidence reveals a gap, ICM can compare training, staffing, governance, integration, and recovery improvements under its actual budget and operating constraints. Proposed changes will be tested on a limited scale, with clear responsibilities and a rollback path. We recommend checking current service conditions daily, reviewing performance monthly, and reassessing the improvement plan quarterly.

Carriers can participate through agreed shipment identifiers, compatible status definitions, and jointly confirmed timestamps. These shared measures should help the port and its customers identify where a handoff failed and resolve discrepancies without exchanging unnecessary confidential records.

This study has not inspected ICM's confidential systems. Its numerical examples are simulations, so we cannot certify the company's present maturity or promise a particular improvement. The proposed approach earns confidence through evidence: disclose what has been measured, acknowledge what remains uncertain, and track whether corrective actions work.

We recommend that ICM publish an initial measured baseline after an independently reviewed pilot and provide customers with a clear route for reporting inaccurate or delayed information. Your feedback should become part of the same tracked correction process.

Sincerely,

The D&A assessment consulting team

References

- [1] COMAP. 2022 ICM Problem D: Data Paralysis? Use Our Analysis! 2022. Problem statement supplied with this project, pp. 1–3.
- [2] I. de la Peña Zarzuelo, M. J. Freire Soeane, and B. López Bermúdez. Industry 4.0 in the port and maritime industry: A literature review. *Journal of Industrial Information Integration*, 20:100173, 2020. <https://doi.org/10.1016/j.jii.2020.100173>.
- [3] R. Albertoni and A. Isaac, eds. *Data on the Web Best Practices: Data Quality Vocabulary*. W3C Working Group Note, 15 December 2016. <https://www.w3.org/TR/vocab-dqv/>.
- [4] P. Groth and L. Moreau, eds. *PROV-Overview: An Overview of the PROV Family of Documents*. W3C Working Group Note, 30 April 2013. <https://www.w3.org/TR/prov-overview/>.
- [5] S. Särkkä. *Bayesian Filtering and Smoothing*. Cambridge University Press, 2013. <https://doi.org/10.1017/CB09781139344203>.
- [6] T. Gneiting and A. E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007. <https://doi.org/10.1198/016214506000001437>.
- [7] D. Bertsimas and M. Sim. The price of robustness. *Operations Research*, 52(1):35–53, 2004. <https://doi.org/10.1287/opre.1030.0065>.
- [8] National Institute of Standards and Technology. *Framework for Improving Critical Infrastructure Cybersecurity*, Version 1.1, 2018. <https://doi.org/10.6028/NIST.CSWP.04162018>.

Source-use boundary. References support concepts and methods, not the invented numerical test parameters. The NIST citation refers to the historical version appropriate to the 2022 case; it is not a claim that Version 1.1 is the latest framework. Public references were checked during preparation on September 19, 2026.

Layout reference. The user supplied the 25-page 2023 ICM D paper by Team 2300229, titled *Methods of Measuring Priority via Graph Models: Who Is the Top 1?* We used its summary-sheet arrangement, numbered hierarchy, running header, and figure/table conventions as visual references. Its topic, models, data, figures, and team number were not reused as this study’s content.

20 Reproducibility and AI-use disclosure

20.1 Reproduce the results

The accompanying archive contains the complete Python implementation, synthetic event file, pilot-profile file, KPI counts, result tables, vector figures, and both LaTeX entry points. Running `python code/run_models.py` regenerates the experiments with NumPy seed 2022. The generator executes before every fit, so source observations and generated truth remain distinguishable. Python, NumPy, SciPy, pandas, and Matplotlib are required; installed versions are recorded in `code/requirements.txt`.

The state generator uses persistence 0.88, action coefficient 0.24, baseline logits $(-0.65, 0.05, -0.90)$, and onset delays $(2, 1, 2)$. Actions follow four active months in a twelve-month cycle, offset by dimension. Unknown outcomes occur at probability 0.025 except failures in months 40–41, where it is 0.20. Three percent duplicate rows are appended; late flags have probability 0.025. The chain-generator coefficients, ordered as intercept, P, T, R, $P \times R$, workload, and outage, are

$$\begin{pmatrix} -2.8 & 1.4 & 1.6 & 1.8 & .8 & -.5 & -1.4 \\ -3.0 & 1.7 & 1.3 & 2.0 & .6 & -.7 & -1.7 \end{pmatrix}. \quad (24)$$

These coefficients belong only to the generator. Fitted coefficients are exported separately in `results/metrics.json`. The optimizer consumes fitted probabilities, not the generating coefficients. Its stored outputs identify the exact project menu and all evaluated baselines.

20.2 Compile the documents

Compile each entry point twice with XeLaTeX, or use the supplied build script. The shared source selects English or Chinese text, preserving identical equations, figures, tables, and numerical macros. Vector figures are already included; Python is not required merely to compile the PDFs. Team number is controlled by one command in `preamble.tex`.

20.3 AI-use disclosure

This retrospective study was prepared with OpenAI Codex, an AI assistant, on September 19, 2026. The user supplied the modeling framework, the 2022 problem, and the 2023 layout example. The request was to analyze the framework, complete the solution, produce Chinese and English PDFs, and package the LaTeX sources with an English limit of 25 pages.

AI assistance covered model criticism, literature lookup, simulation and optimization code, bilingual drafting, LaTeX production, and document checks. Numerical claims were generated by executed code and checked against saved outputs. The report includes unsuccessful validation results rather than concealing them. No human expert interviews, corporate access, external certification, or independent peer review occurred. This disclosure describes the present preparation process and is included in the total page count.